

Senior Design Project Review

Interpretable AI System for Parkinson's Disease and Neurological Disorder Prediction using Machine Learning and Quantum Models

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Presented By:

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SCOPE

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Presentation Outline

- Introduction
 - Project Overview and Problem Statement
 - Motivations
- Background & Related Work/ Literature Review
 - Existing Solutions / Literature Survey
 - Gaps Identified / Improvements Over Existing Solutions
- Project Objectives
- Proposed Solution
 - System Architecture /Workflow Diagram /Model Diagram /Block Diagram/UML diagrams
 - Key Components/Features/Modules
 - Algorithms Used/Technologies, Frameworks, and Tools Used
- Simulation and Results
- Summary (Findings, Limitations, Future Directions)
 - References

Project Overview

- Parkinson's Disease is a progressive neurological disorder that affects movement, speech, and motor control. Speech impairment is one of the commonly observed symptoms in Parkinson's patients, where changes may appear in voice stability, pitch variation, loudness, and signal complexity
- This project focuses on predicting Parkinson's Disease using numerical speech features extracted from voice recordings. Instead of using the complete high-dimensional dataset, the system selects the top 10 important speech features and trains multiple machine learning models for classification
- Along with classical models such as SVM, XGBoost, PSO-SVM, this project also explores hybrid quantum machine learning models such as QSVM and VQC. These models combine classical preprocessing with quantum-inspired or quantum circuit-based learning approaches, allowing us to compare traditional machine learning with emerging quantum-based methods

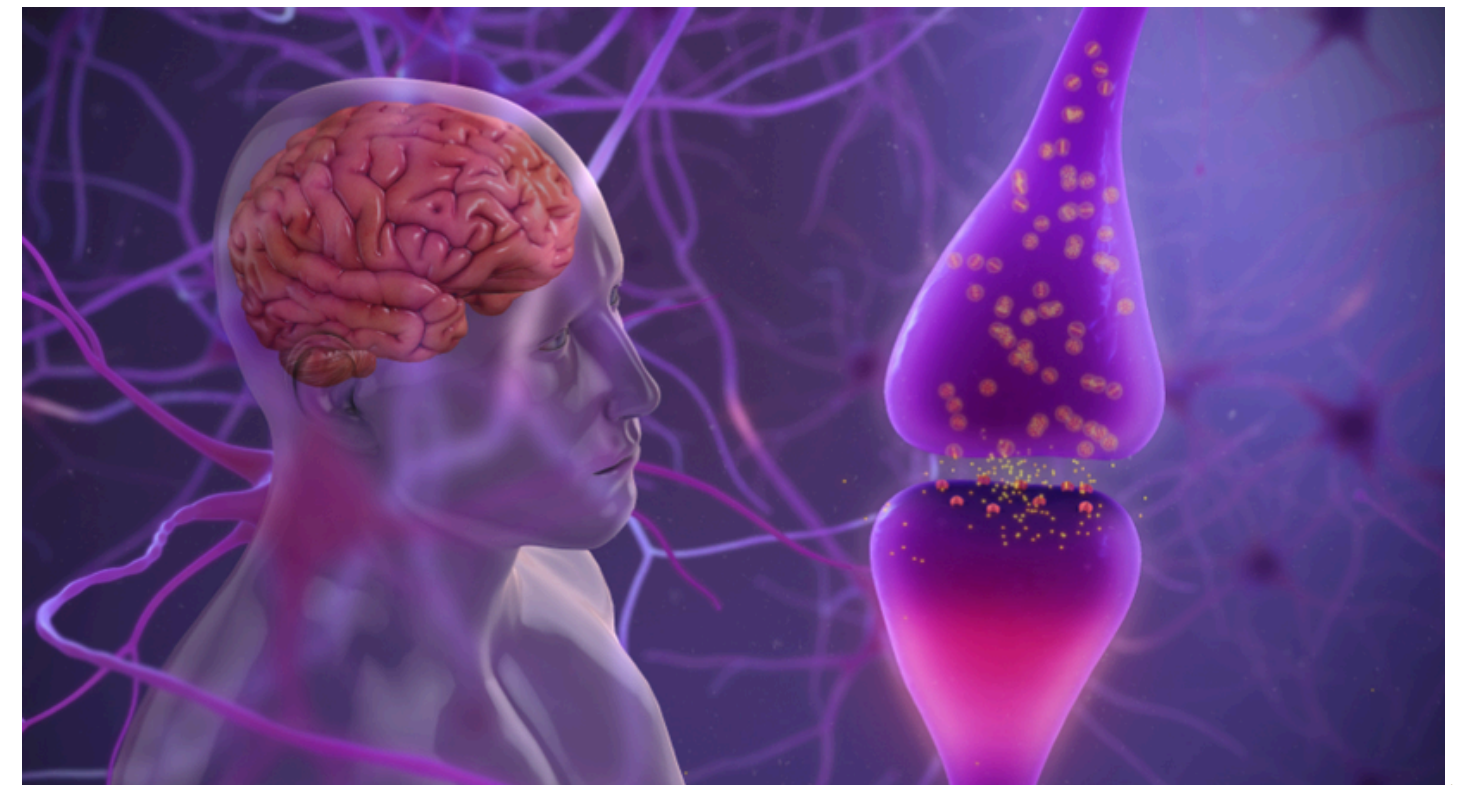


Problem Statement

- The project aims to develop an efficient and interpretable prediction system using speech-derived acoustic features to classify whether a voice sample corresponds to a Healthy Control or Parkinson's Disease pattern. Since speech impairments such as reduced vocal intensity, monotonic tone, voice tremors, and irregular speech patterns can appear in the early stages of Parkinson's Disease, speech-based analysis can support non-invasive early screening
- The original PD Speech Features Dataset contains a large number of vocal measurements, which may increase model complexity and reduce interpretability. Therefore, feature selection techniques are used to identify the most informative top speech features for prediction. The system compares classical machine learning models, hybrid quantum machine learning models, and a swarm-optimized model to evaluate their effectiveness in Parkinson's voice classification
- The project also focuses on explainability by showing how important speech feature groups influence the prediction result. The final system provides Parkinson's probability, classification output, confidence score, and interpretable visual explanations through an interactive web application

Motivations

- Parkinson's Disease affects millions of people worldwide and early screening is important for timely medical attention. Since voice changes are often associated with Parkinson's symptoms, speech analysis provides a non-invasive and cost-effective way to support screening research
- A major motivation of this project is to reduce model complexity. The dataset contains many speech features, but not all of them contribute equally to prediction. By selecting only the top 10 important features, the system becomes simpler, faster, and easier to explain
- Another motivation is to compare both classical and hybrid quantum models. Classical models are well-established and reliable, while hybrid quantum models are emerging approaches that may offer new possibilities for classification tasks. Including both allows us to study their performance under the same dataset and feature conditions



Background & Related Work / Literature Review

Existing Solutions / Literature Survey

- Several studies have used speech signal features for Parkinson's disease detection
- Machine learning models such as SVM, Random Forest, Decision Tree, KNN, and Logistic Regression have been used for classification
- Deep learning approaches have also been explored for voice and biomedical signal analysis
- Existing research shows that features such as jitter, shimmer, pitch, frequency variation, and noise-to-harmonic ratio are useful for identifying speech abnormalities
- Some systems focus mainly on prediction accuracy, while fewer systems provide clear explanation of the prediction

Literature Survey

S.No	Paper Title	Year	Published In	Proposed Methodology	Advantage	Disadvantage
1	Speech biomarkers of Parkinson's disease: Advances and open challenges Author: Vasquez-Correa et al.	2024	Biomedical Signal Processing and Control	Reviewed speech biomarkers such as jitter, shimmer, HNR, DFA, RPDE, and PPE for Parkinson's disease analysis.	Clearly explains important speech biomarkers linked with motor dysfunction.	Mainly a review study; does not provide a complete implementation model.
2	Remote speech analysis for monitoring Parkinson's disease progression Author: Tsanas et al.	2024	IEEE Reviews in Biomedical Engineering	Used remote speech analysis and sustained phonation tasks to monitor Parkinson's disease progression.	Supports non-invasive and remote monitoring through speech signals.	Performance depends on recording quality and controlled speaking conditions.
3	Speech-based biomarkers for early Parkinson's disease detection: Advances in acoustic analysis and machine learning Author: Xu et al.	2025	IEEE Transactions on Neural Systems and Rehabilitation Engineering	Applied acoustic feature extraction with ML models such as SVM, Random Forest, and boosting methods.	Achieved strong performance using acoustic biomarkers for early detection.	Model reliability depends on dataset size, feature quality, and generalization.
4	Transformer-based speech representation learning for Parkinson's disease detection Author: SedighMalekroodi et al.	2025	Computer Speech & Language	Used transformer-based speech representation learning for Parkinson's disease classification.	Learns complex temporal and spectral speech patterns automatically.	Requires large datasets and high computational resources.
5	Quantum machine learning for clinical decision support systems: Opportunities and limitations Author: Chow	2025	Nature Digital Medicine	Reviewed quantum machine learning for clinical decision support systems.	Highlights future potential of QML in healthcare applications.	Current quantum systems have qubit limitations and noise-related issues.

Background & Related Work / Literature Review

Gaps Identified / Improvements over Existing Solutions

Gaps Identified:

- Many existing models act like black-box systems
- Limited focus on explainability and feature importance
- Some systems do not provide a user-friendly interface
- Many solutions focus only on accuracy and ignore interpretability

Improvements in Our Project:

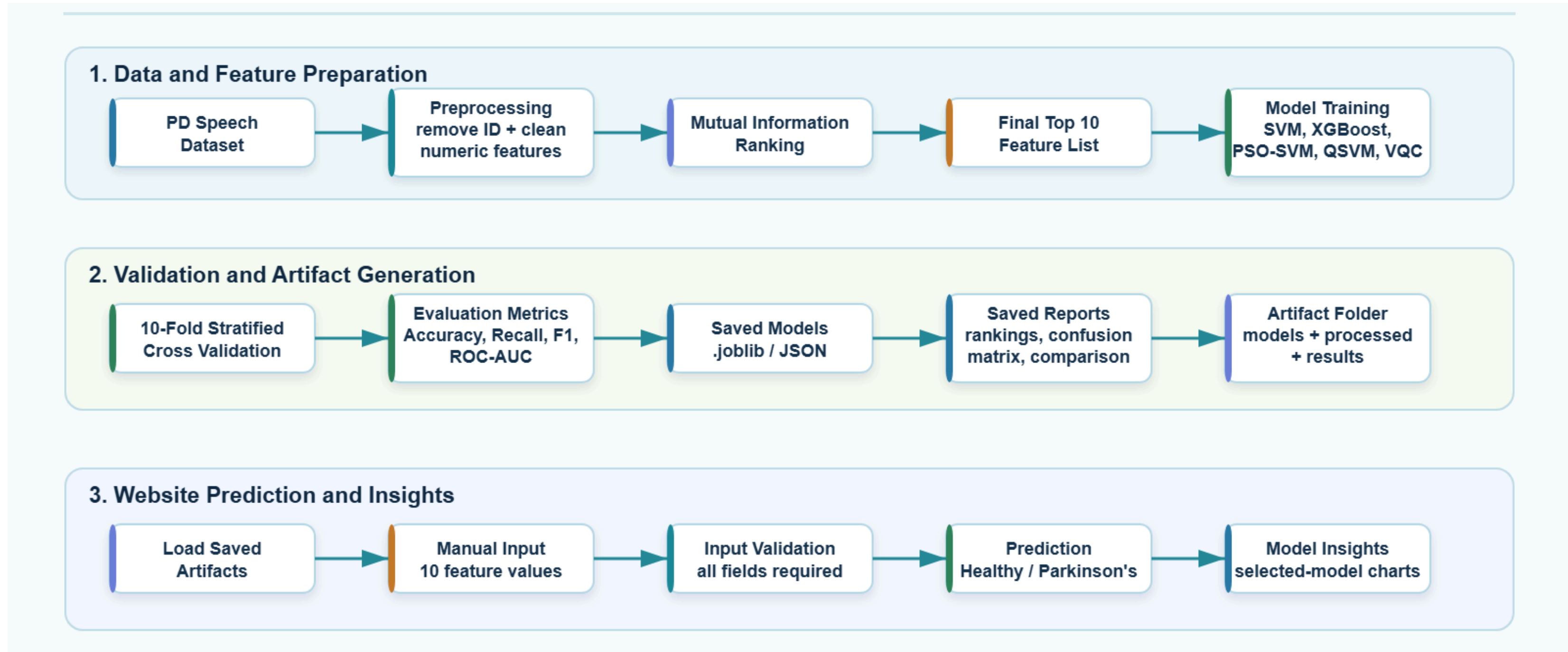
- Uses Explainable AI to show important speech features
- Provides prediction results through a dashboard
- Compares multiple machine learning and Quantum models
- Focuses on both performance and interpretability
- Presents results using charts, tables, and visual explanations

Project Objectives

- Analyze speech-based characteristics associated with Parkinson's disease
 - Preprocess and refine the dataset for effective model training
 - Build and train machine learning models for classification tasks
 - Evaluate and compare model performance using relevant metrics
 - Identify key features that significantly influence predictions
 - Incorporate Explainable AI techniques to improve transparency
- Design a dashboard for visualizing predictions and model explanations

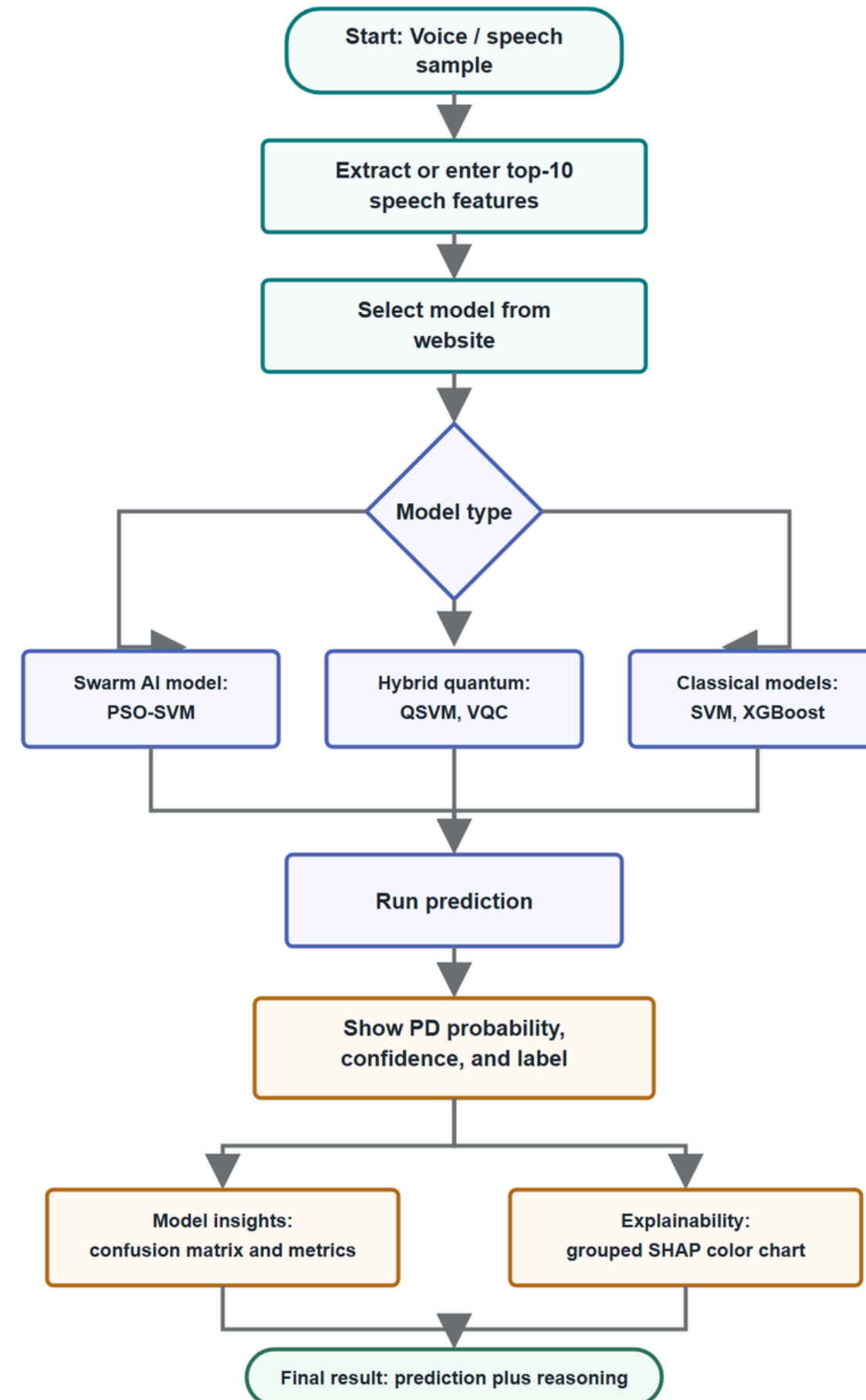
Proposed Solution

- **System Architecture**



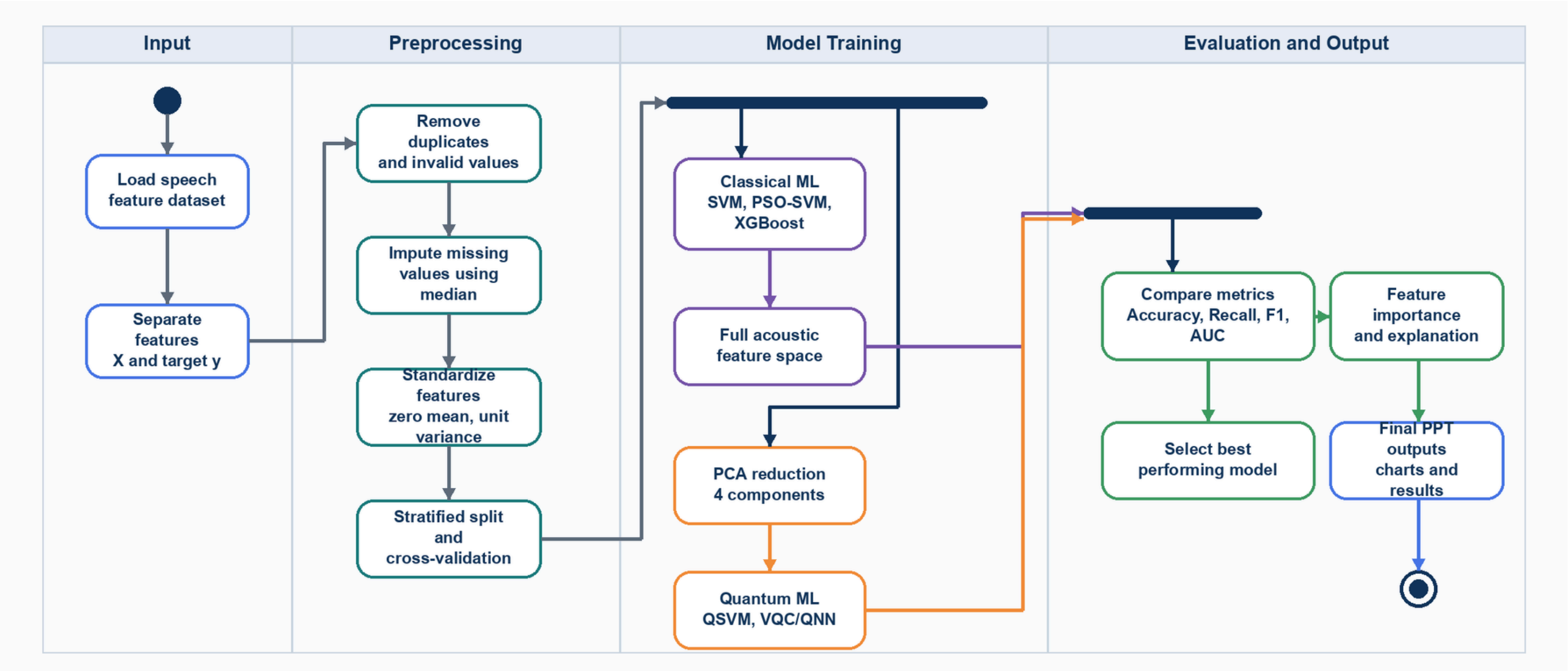
Proposed Solution

Workflow



Proposed Solution

UML diagrams



Key Components

Application Entrypoint

- FastAPI backend initializes and serves the prediction application

Backend Application Layer

- Handles model loading, prediction requests, validation, and API responses
- Manages trained model and result files

Model Layer

- Stores trained SVM, XGBoost, QSVM, VQC, and PSO-SVM models
- Includes model metrics, confusion matrices, feature importance

Frontend Layer

- React and TypeScript interface for model selection, input entry, prediction, and visual results
- Displays SHAP charts, model insights, recent predictions

Proposed Solution

Features

1. Top-10 Speech Feature Engineering

- Uses selected top-10 speech features from the PD voice dataset
- Includes energy, frequency, entropy, and wave-pattern based features

2. Prediction Models

- Supports SVM, XGBoost, QSVM, VQC, and PSO-SVM
- Predicts whether the voice sample is Parkinson's or Healthy Control

3. Explainability Using SHAP

- Shows grouped SHAP-based feature contribution analysis
- Identifies which speech feature groups increased or lowered the score

4. Interactive Web Interface

- User-friendly React-based interface
- Displays prediction result, confidence, probability, and model name

5. Model Insights

- Shows confusion matrix, model metrics, and feature importance
- Helps users understand model behavior and prediction reasoning

Algorithms

- SVM → Classical Parkinson's voice classification
- XGBoost → High-performance prediction and feature importance analysis
- QSVM → Quantum kernel-based classification approach
- VQC → Hybrid quantum circuit-based classification
- PSO-SVM → Swarm-optimized SVM for improved hyperparameter tuning
- SHAP → Model explainability and feature contribution analysis
- Top-10 Feature Selection → Reduced speech feature representation

Technologies & Frameworks

- Python for machine learning, model training, and backend development
- React and TypeScript for frontend website development
- FastAPI for backend APIs and prediction request handling
- Pandas and NumPy for preprocessing and data handling
- Scikit-learn and XGBoost for classical machine learning workflows
- Qiskit for quantum machine learning models
- Particle Swarm Optimization for swarm-based model tuning
- Recharts for charts, explainability graphs, and model insights
- Joblib for saving and loading trained model files

Development Tools

- Google Colab and Jupyter Notebook for experimentation and model training
- Visual Studio Code for application development
- Git and GitHub for version control and project management
- Python Virtual Environment for dependency management

System Modules

- Validation Module for checking the 10 input speech feature values
- Feature Processing Module for preparing selected top-10 model inputs
- Prediction Module for Parkinson's voice classification
- Explainability Module using SHAP for grouped feature contribution analysis
- Model Insights Module for confusion matrix, metrics, and feature importance

- **Dataset Descriptions / Input – Output Descriptions**

- The dataset contains speech-based biomedical voice measurements
- Each record represents extracted voice features from an individual
- Features include frequency-based, amplitude-based, and noise-related speech measurements
- The target variable indicates whether the speech sample belongs to a Parkinson's disease case or a healthy control case

Input:

- Speech feature values such as jitter, shimmer, pitch, frequency measures, and noise ratio

Output:

- Predicted class label along with important features influencing the result

Parameters Descriptions (if any) / Initial Conditions

- Dataset: pd_speech_features.csv
- Target column: class
- Feature selection method: Mutual Information
- Selected feature count: 10
- Cross-validation: 10-fold stratified cross-validation
- Feature mode: direct_top10
- Prediction threshold: 0.5
- Models trained: SVM, QSVM, VQC, XGBoost, PSO-SVM
- Evaluation metrics: Accuracy, Recall, F1-score, ROC-AUC, Confusion Matrix
- Website default model: XGBoost
- For quantum models, training is more computationally expensive. Therefore, QSVM and VQC may use a stratified subsample strategy where required, while classical models are trained on the full dataset

- Implementation Outcomes (Tables / Figures / User Interface Snaps)

Top-10 PD
Speech Model Lab

Home

Input

Prediction

Explainability

Model Insights

About

Researcher

v0.1.0

Modelpd_speech_features_local_top10_mi_XGBoostModel StatusInference-ReadyBackendClassicalFeature Count10

Top-10 Feature PD Speech Model Lab

Enter the 10 selected speech-feature values, run the reduced-feature PD models, and review the grouped reasoning behind each result.

Start AnalysisView Models

Enter Feature Values

Use the 10 numeric speech-feature values expected by the trained model.

Go to Input

Receive Predictions

Review classification, probability, confidence, source row, and model.

Open Prediction

Understand The Reasoning

Use grouped SHAP explanations and the local XAI chatbot.

Open Explainability

Datasets1

Inference-ready models5

Selected features10

Selected familyClassical

Predictions are for research support only and are not intended for clinical decision-making or diagnosis.

Home

Input

Prediction

Explainability

Model Insights

About

Researcher

v0.1.0

Enter The 10 Speech Feature Values

Manual single-case prediction, no CSV or raw audio required.

Example Row1Use Sample ValuesClear

Top-10 Feature Form

Use sample values for demonstration, then replace them with calculated feature values for a new case.

1

Sudden loudness-change variation

std_delta_delta_log_energy

Measures how much the acceleration of voice energy changes across the recording.

standard deviation

0.016392

2

Average voice energy bursts

tqwt_TKEO_mean_dec_12

Represents the typical rapid energy level in one wave-pattern band.

TKEO mean

0.0000233

3

Rapid tone-change variation

std_8th_delta_delta

Captures fast changes in one tone/resonance pattern of the voice.

standard deviation

0.012829

4

Average rapid voice energy

tqwt_TKEO_mean_dec_16

Represents the typical strength of quick energy bursts in a mid-detail signal band.

TKEO mean

0.008643

5

Fine voice irregularity

tqwt_entropy_log_dec_35

Shows how uneven the very small vibration patterns are in the voice signal.

log entropy score

-3190.1752

6

Voice energy burst consistency

tqwt_TKEO_std_dec_12

Shows how much quick energy bursts vary in one signal band.

TKEO standard deviation

0.000059

7

Fine voice disorder

tqwt_entropy_shannon_dec_35

Measures how spread out or unpredictable tiny voice vibrations are.

Shannon entropy

0.015562

8

Average vocal tone pattern

mean_MFCC_2nd_coef

Summarizes one of the main sound-shape patterns related to vocal tract resonance.

MFCC average

2.4874

9

Lower-band voice irregularity

tqwt_entropy_log_dec_11

Shows unevenness in a broader voice wave-pattern band.

log entropy score

-229943.2967

10

Strongest voice vibration peak

tqwt_maxValue_dec_12

The highest signal peak found in one wave-pattern band.

maximum band value

0.024286

pd_speech_features_local_top10_mi_XGBoostAll 10 values ready

Predict From Values

Predictions are for research support only and are not intended for clinical decision-making or diagnosis.



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Simulation and Results

Implementation Outcomes

Speech Model Lab

Home

Input

Prediction

Explainability

Model Insights

About

Model

pd_speech_features_local_top10_mi_XGBoost

Model Status

Inference-Ready

Backend

Classical

Feature Count

10

New Input

CLASSIFICATION

Parkinson's (PD)

Source sample-edited, row 1

Probability

95.9%

Confidence

95.9%

Model

XGBoost

Review the reasoning

The explainability page shows grouped SHAP contributions and a local chatbot for this result.

Open Explainability →

Recent Predictions

Latest local results from CSV or edited sample rows.

Source	Row	PD Probability	Predicted Label	Model
sample-edited	1	0.959	Parkinson's (PD)	pd_speech_features_local_top10_mi_XGBoost

Download CSV

Predictions are for research support only and are not intended for clinical decision-making or diagnosis.

Model

pd_speech_features_local_top10_mi_XGBoost

Model Status

Inference-Ready

Backend

Classical

Feature Count

10

Run New Prediction

Explainability

Visualize grouped SHAP contributions and ask local questions about the latest prediction.

Native SHAP

GROUPED EXPLANATION

What most influenced the score

Red bars raised the displayed probability. Green bars lowered it from the model baseline.

Wave-pattern complexity

Energy variation

Frequency pattern shifts

-3 pp

-2 pp

0 pp

2 pp

3 pp

HIGH RISK

Parkinson's Positive

High confidence using XGBoost

Probability

95.9%

Confidence

95.9%

Baseline

96.5%

Grouped Shift

-0.59 pp

Main reasons the score increased

Energy variation

4 grouped features

Energy variation moved this sample toward a Parkinson's-positive prediction.

+1.86 pp

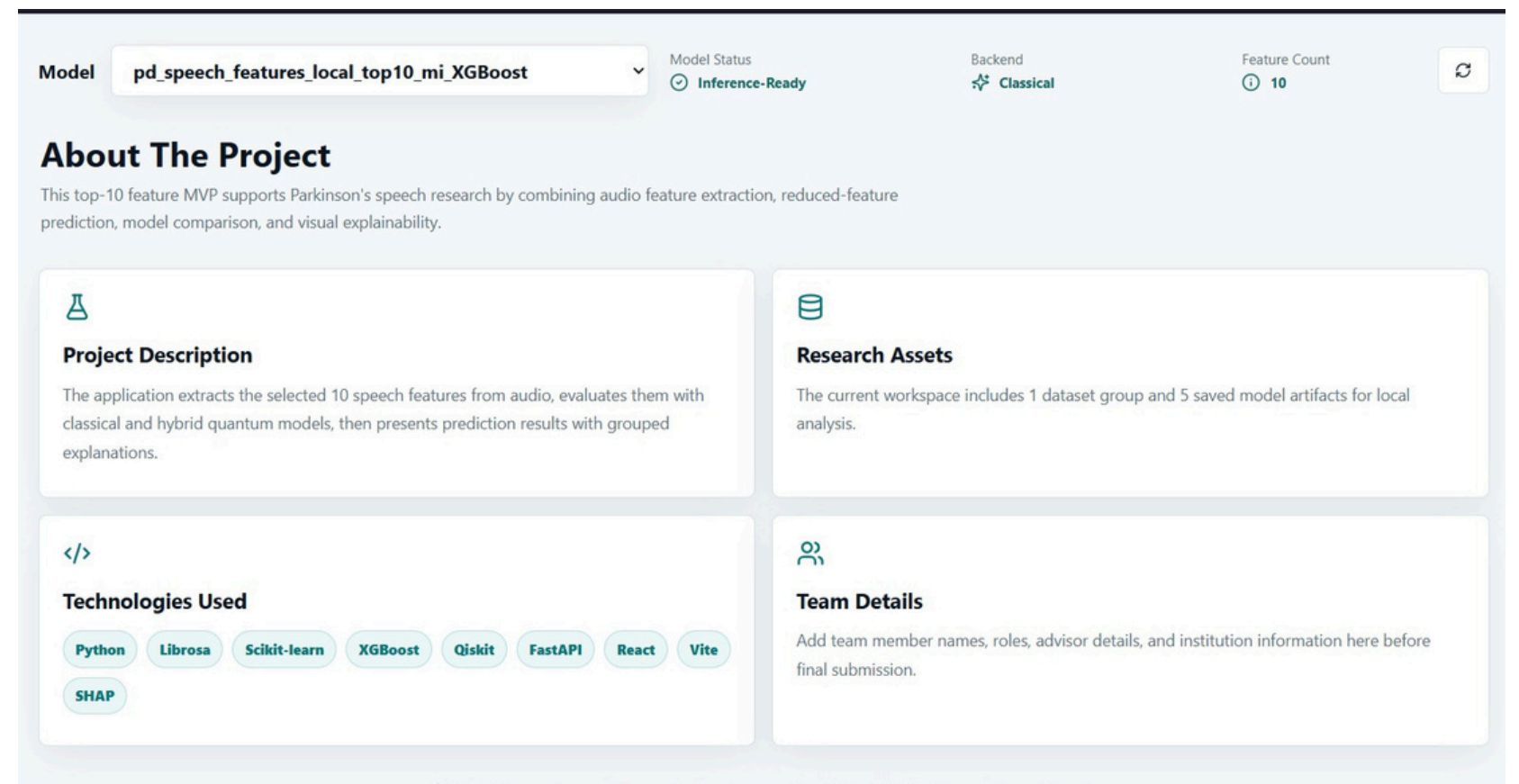
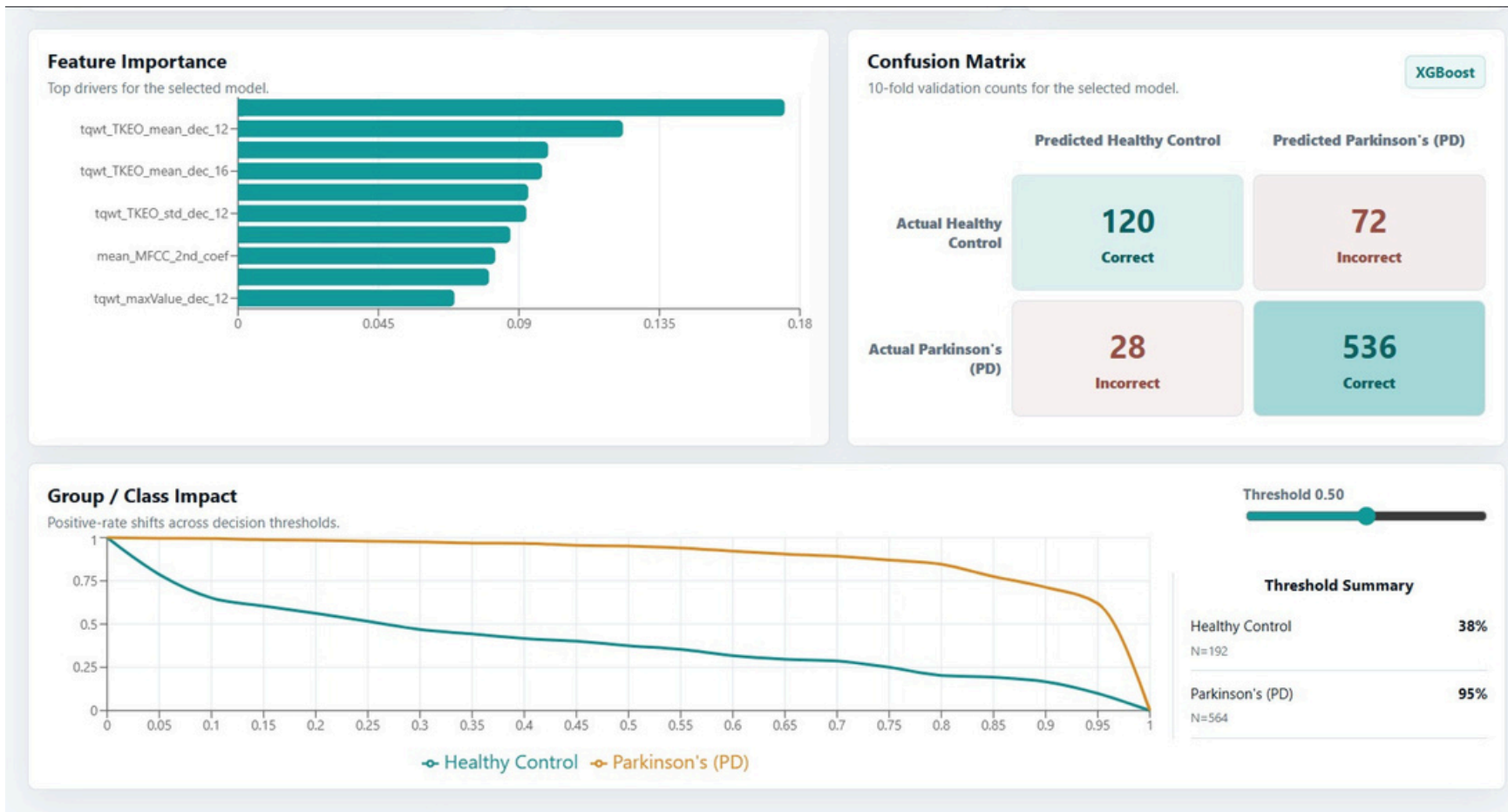
Frequency pattern shifts

2 grouped features

Frequency pattern shifts moved this sample toward a Parkinson's-positive prediction.

+0.45 pp

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Key Findings

- Top-10 selected speech features can effectively support Parkinson's voice classification
- XGBoost and SVM provide strong classical ML performance
- QSVM and VQC show the possibility of hybrid quantum ML comparison
- PSO-SVM adds swarm-based optimization for better SVM parameter tuning
- SHAP-based explainability helps identify which voice feature groups influence predictions.

Limitations

- Performance depends on the quality and diversity of the available voice dataset
- Quantum models are currently simulated and may be limited by system resources
- Manual feature input requires correct extraction of the same 10 speech features

Future Directions

- Add direct audio upload with automatic feature extraction for all workflows
- Improve quantum and swarm models with deeper tuning and comparison
- Add more explainability views for doctors, researchers, and patients
- Deploy the system as a secure web application for broader testing

APA format :

- Hossain, M. A., & Amenta, F. (2024). Machine learning-based classification of Parkinson's disease patients using speech biomarkers. *Journal of Parkinson's Disease*, 14(1), 95–109.
<https://doi.org/10.3233/JPD-230002>
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<https://doi.org/10.1016/j.jvoice.2024.05.018>
- Di Cesare, M. G., Perpetuini, D., Cardone, D., & Merla, A. (2024). Machine learning-assisted speech analysis for early detection of Parkinson's disease: A study on speaker diarization and classification techniques. *Sensors*, 24(5), 1499. <https://doi.org/10.3390/s24051499>
- Jeong, S.-M., Song, Y.-D., Seok, C.-L., Lee, J.-Y., Lee, E. C., & Kim, H.-J. (2024). Machine learning-based classification of Parkinson's disease using acoustic features: Insights from multilingual speech tasks. *Computers in Biology and Medicine*, 182, 109078.
<https://doi.org/10.1016/j.compbiomed.2024.109078>
- Tougui, I., Zakroum, M., Karrakchou, O., & Ghogho, M. (2024). Transformer-based transfer learning on self-reported voice recordings for Parkinson's disease diagnosis. *Scientific Reports*, 14, 30131.
<https://doi.org/10.1038/s41598-024-81824-x>

Thank you

Any question?